

Functional & Performance Testing Template

Model Performance Test

Date	6 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Test Scenarios & Results:

Functional Testing:

Test Case ID	Scenario (What to Test)	Test Steps	Expected Result	Actual Result	Pass/Fail
FT-01	Soil Parameter Input Validation	Enter valid and invalid N, P, K, and pH values	Valid values accepted, invalid values rejected	The application accepted valid soil parameter values and prevented invalid or incomplete inputs from proceeding to prediction.	Pass
FT-02	Environmental Parameter Input Validation	Enter Temperature, Humidity, and Rainfall values	Valid values accepted	The application successfully accepted valid environmental parameter values and displayed validation messages for invalid inputs.	Pass

FT-03	Crop Recommendation Generation	Enter all required parameters and click Predict	Correct crop recommendation displayed	The trained machine learning model generated a crop recommendation based on the provided inputs, and the result was displayed successfully.	Pass
FT-04	Machine Learning Model Loading	Launch the Flask application	Trained .pkl model loads successfully	The Flask application successfully loaded the trained machine learning model (.pkl) during startup without errors.	Pass
FT-05	Result Display	Submit valid inputs	Recommended crop displayed correctly	The recommended crop was displayed clearly on the results page after prediction.	Pass

Performance Testing:

PT-01	Prediction Response Time	Submit a prediction request	Prediction generated within 3 seconds	Average prediction response time was approximately 1.2 seconds , meeting the expected performance.	Pass
PT-02	Multiple Prediction Requests	Perform several prediction requests consecutively	Application remains responsive	The application handled multiple prediction requests without noticeable delay or failure.	Pass
PT-03	Application Stability	Run the application continuously during testing	No crashes or unexpected errors	The application remained stable throughout testing and completed all prediction requests successfully without crashes.	Pass

Testing Summary:

Metric	Result
Total Test Cases	8
Passed	8
Failed	0
Success Rate	100%